Routing Algorithms:- This is a part of network layer software responsible for deciding which output line an incoming packet should be transmitted on.

Types of Routing Algorithms
→ Non-adaptive
→ Adaptive (Dynamic Routing.

Non-Adaptive (Static Routing)
* Routing decisions does not depend on measurements of current topology.

Adaptive (Dynamic Routing).
* It is dependent on the measurements of current topology.

Routing Algorithms
1) Optimality Principle
* We should find the best route according to the path cost

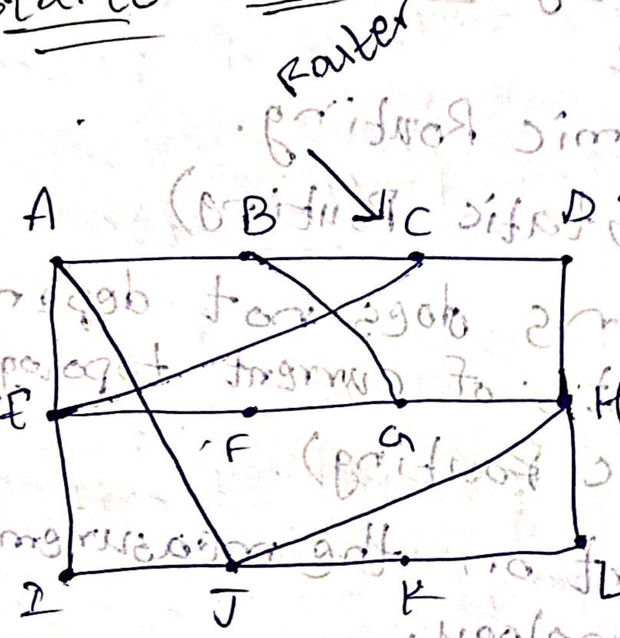
2) Flooding
* Network information like topology, load condition, cost of different paths are not required.
* Every incoming packet to a node is sent out on every outgoing line except the one it arrived on.
* This generates vast number of duplicate packets.

3.) **Shortest Path Routing**

* The algorithm finds the shortest path from one node to another node in the network.
* This uses Dijkstra's Algorithm.

Distance Vector Routing

14/02/26



8	A
20	A
28	I
20	H
17	I
30	I
18	H
12	H
10	I
0	-
6	K
15	K

JB → JA + AB = 8 + 12 = 20

JC → JI + IC = 10 + 18 = 28

JD → JH + HD = 8 + 12 = 20

JE → JI + IE = 10 + 7 = 17

JF → JI + IF = 10 + 20 = 30

JG → JH + HG = 12 + 6 = 18

JH → JH + HH = 12

JI → JI = 10

JJ → 0

JK → 6

JL → JK + KL = 6 + 9 = 15

Count to infinity

* Drawback of the distance vector routing.

A	B	C	D	E	
•	•	•	•	•	Initially
	1	•	•	•	After 1 exchange
	1	2	•	•	After 2 exchange
	1	2	3	•	After 3 exchange
	1	2	3	4	After 4 exchange.

* Paritcularly reacts rapidly to good news, but leisurely to bad news.

Link State Routing

Five steps.

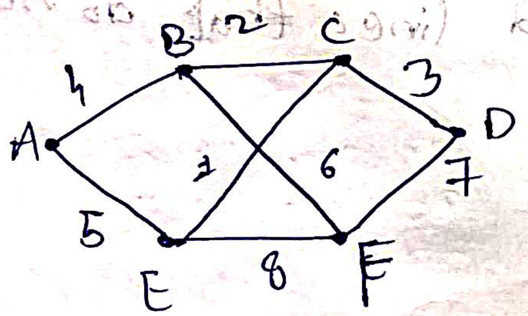
1.) Discover it's neighbors, learn their network address.

2.) measure the delay or cost to each of it's neighbors.

3.) Construct a packet telling all it has just learned.

4.) Send this packet to all other routers.

5.) Compute the shortest path to every other router.



A	
B	4
E	5

B	
C	2
A	4
F	6

C	
B	2
D	3
E	1

D	
C	3
F	7

E	
A	5
C	1
F	8

F	
B	6
D	7
E	8

~~Hierar~~ Broadcast Routing

* Sending packets to all destinations simultaneously.

Methods

1) Forwarding a packet from source to destination.
2) Flooding
3) multi destination routing.
4) spanning tree.
5) Reverse path forwarding.

~~Hierarchial Routing~~

Multicast Routing

* Routers must update information like who is added to a group, who leaves the group & which host belongs to which group.

→ Routers must construct a spanning tree covering all other routers.

→ When a process sends a multi cast packet to a group, the first router examines its spanning tree & Prunes it. (removes all unwanted lines that do not lead to host).

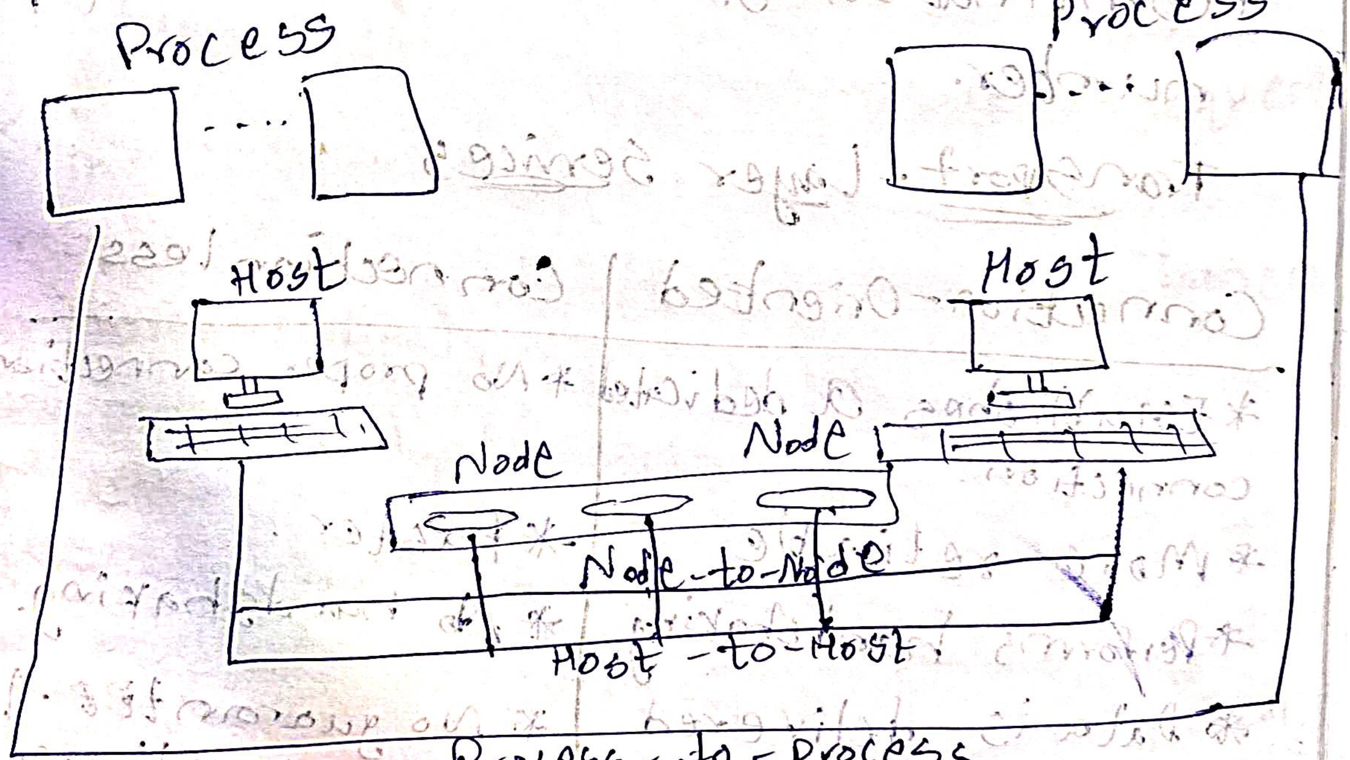
Process to Process Delivery

21/02/26

* The data link layer is responsible for delivery of frames b/wn two neighbouring nodes.
* The network layer is responsible for the delivery of datagrams b/wn two hosts. (Host to Host delivery).
* Transport layer is responsible for Process to Process delivery.

Types of Data Delivery

→ Node-to-Node (Data link layer)
→ Host-to-Host (Network layer)
→ Process-to-Process. (Transport layer).



Client/Server Paradigm

* Process-to-Process communication is done by client/server paradigm.
* Done by,

→ Local host → Remote host
→ Local process → Remote process.

Addressing

* In transport layer, the source & destination processes are identified by a port number.
* IP addresses & port numbers play different roles
* Destination IP addresses defines the host among all the hosts in the world.

Socket Address

* Combination of a IP address & port number.

Transport Layer Services

Connection-Oriented	Connectionless
* Establishes a dedicated connection.	* No proper connections.
* More reliable	* Faster.
* Performs handshaking.	* No handshaking.
* Data is delivered in the same order.	* No guarantee that the data is delivered in the same order.

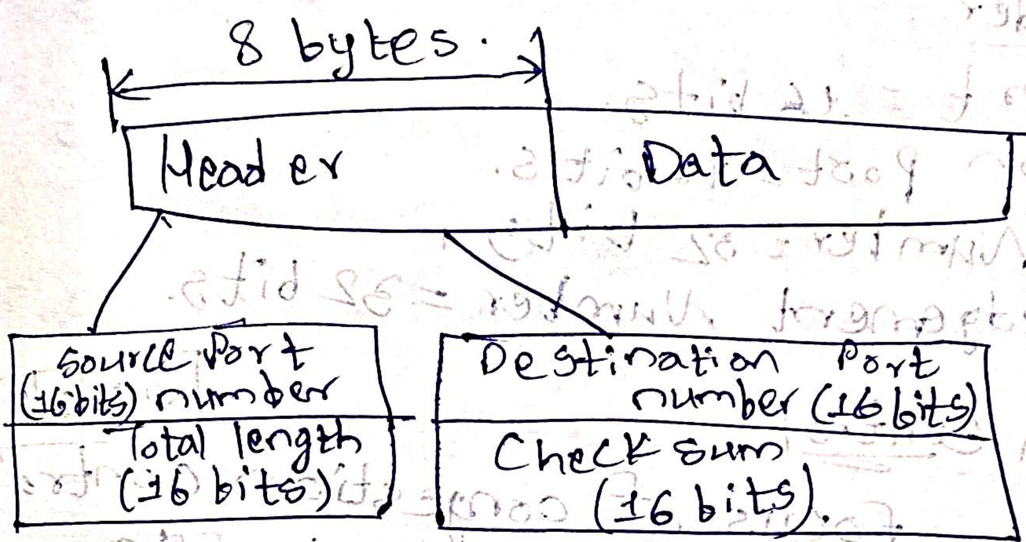
Protocols

→ SCTP (Stream Control Transmission)
→ TCP (Transmission Control)
→ UDP (User Datagram Protocol)

Protocols.

User Datagram Protocol

* This is a connectionless unreliable transport protocol.
* UDP header is an 8 byte fixed & simple header.

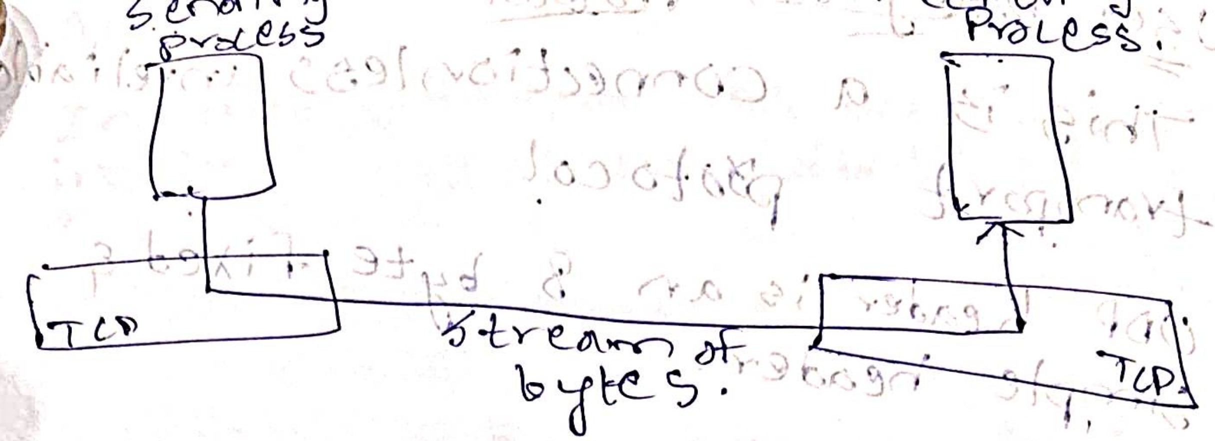


Uses

* This is suitable for a simple request response.
* UDP file is used in Trivial File Transfer Protocol (TFTP).
* Used in multicasting.
* Used in management process such as SNMP.

TCP (Transmission Control Protocol)

* This is a connection oriented protocol.
* Creates a virtual connection b/w two TCP's to send data.
* Uses Internet flow & error control mechanisms at the transport layer.



TCP Header

→ Source Port = 16 bits.
→ Destination Port = 16 bits.
→ Sequence Number = 32 bits.
→ Acknowledgement Number = 32 bits.

Congestion Control 27/2/26

* The main focus of congestion control & quality of service is data traffic.
* Avoids traffic congestion.

Three traffic profiles
→ Constant Bit Rate ⇒ fixed data rate
→ Variable Bit Rate ⇒ rate of data flow changes
→ Bursty ⇒ Data suddenly changes in very short time.

* This may occur if the load on the network...

Congestion Control
- Open Loop
  - Window policy
  - Admission policy
  - Discarding policy.
- Closed Loop
  - Choke packet
  - Hop by Hop choke packet.
  - Explicit congestion notification.

- Retransmission Policy
- Acknowledgement Policy

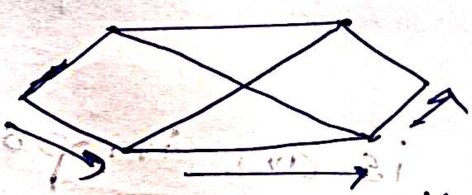
} Open Loop.

Choke Packets (Additional Packet).

* In this, router selects a congested packet & sends choke packet back to the source host, giving it the destination found in the packet.
* When the source host gets the choke packet, it is required to reduce the traffic sent to the specified destination

→ 

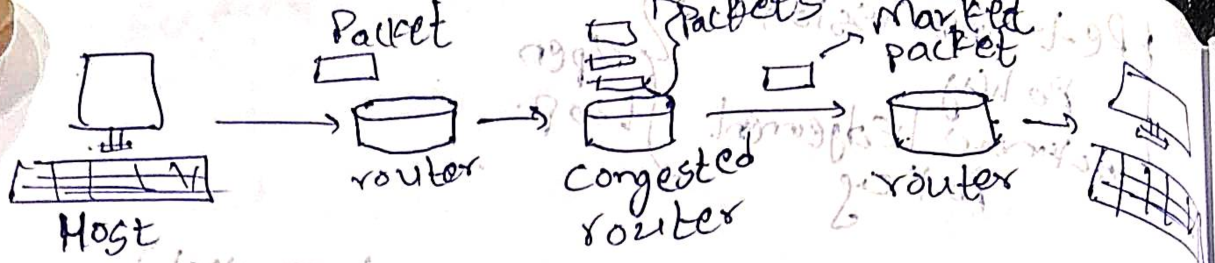
Sending a choke packet.

→ 

Sent Back.

ECN (Explicit Congestion Notification)

* Unlike choke packets a router can tag any packet it forwards ~~bit in~~ ~~the~~ to signal that it is experiencing congestion.
* When the packet is delivered, the destination can note that there is congestion & inform the sender when it sends a reply packet



Hop - by - Hop Backpressure

* At high speeds or over long distances, many new packets may be transmitted after congestion has been signaled because of the delay before the signal takes effect.

Quality of Service

13/03/26

* A stream of packets sent from a source to destination is called a flow.

* Traffic shaping is an important technique which can improve QoS.

→ Over provisioning :- A easy solution which provides more router capacity, buffer space & bandwidth.

→ Buffering :- To provide enough buffer space to store.

28/03/26

Key Parts of the web

1) URL
* Uni-form Resource Locator.
* This is address of a webpage.

Ex:- https://www.kluniversity.com
↓
Domain Name System (DNS).

2) HTTP
* Hyper text Transfer Protocol.
* This is the set of rules that lets your browser & servers to transfer data.

Key Terms
→ HTTP Request
→ HTTP Respone
→ HTTP Status Codes
→ HTTP Headers
→ HTTP Caching
→ HTTP Security
→ HTTPS
→ HTTP/2 & HTTP/3

Status Codes
200 (Success)
301 (Permanent Redirect

HTTP Headers
* These are used for communication between client & the server.
→ General Header ⇒ Applied on request & response headers affecting the without body.

→ Request Header

→ Response Header

→ Entity Header

Headers

→ Age
→ Cache-Control
→ Clear site Data
→ Expires
→ Warnings
→ Pragma

HTTP Caching

* Refers to the mechanism of optimization that the web performance stores HTML pages, images, videos & audio's

DNS

* This is an application layer protocol.
* This is a hierarchial & distributed naming system that translates domain names into IP addresses.
* DNS ensures that the request reaches the correct server by resolving the domain to it's corresponding IP address.

Host Name → DNS → IP-Address

Types

→ Flat name space
→ Hierarchial name space.

Hierarchial name space

* Each name has several parts,
→ 1st part (Nature of organization)
→ 2nd part (name of organization)
→ 3rd part (Department in the organization).

Flat name space

* A name in this space is a sequence of characters without structure

29/3/26

ADHOC Network

An ad Hoc network is a decentralized, wireless network that spontaneously forms when devices with wireless capability come within range of each other.

* ad hoc networks operate without a fixed infrastructure like routers or access points.

Mobile Ad Hoc Networks (MANETs).

* This is a self-configuring, temporary network of mobile devices like smart phones, laptops
* MANET'S are used when fixed infrastructure is unavailable.

Wi-fi

04/04/26

* stands for wireless fidelity.
* Wi-fi is the marketing name from IEEE standard 802.11.
* Uses radio waves that allows devices such as desktop computers, laptops, tablets & cell phones to be able to connect wirelessly to the internet.

Evolution

Wifi 1 (1999)
11B

Wifi 2 (2003)
11G

Wifi 3 (2004)
11A/G

Wifi 4 (2009)
11N

Wifi 5 (2013)
11AC

Wifi 6 (2019)
11AX

Wifi 6E (2021)
11AX

Wifi 7 (2024)
11BE

WiFi 1	2.4GHz	11 Mbps
WiFi 2	5GHz	54 Mbps
WiFi 3	2.4GHz	54 Mbps
:	:	:
WiFi 7	2.4GHz, 5GHz, 6GHz	46Gbps

→ 802.11bb :- Li-Fi, this is the Global communication standard.

* It can deliver faster, more reliable wireless communications with better security compared to Wi-Fi & 5G.

→ 802.11ab (Wi-Fi HaLow).

→ 802.11ay (Next Gen 60 Hz).

Futuristic Wi-Fi Standards

→ Battery-free IoT networks.

→ Augmented / Virtual Reality (AR/VR) & Metaverse.

→ Wi-Fi 8

→ AI & ML.

BLUETOOTH

* This is a PAN protocol designed for communicating over relatively short distances. This is began in 1998 with the launch of the Bluetooth Special Interest Group (SIG).

Features

- → 1 Mbps (data rate)
- → Low power consumption & cost.
- → Ability to handle both voice & data simultaneously.
- → Formation of Piconet (Collection of Bluetooth devices which are synchronized to the same hoping sequence).

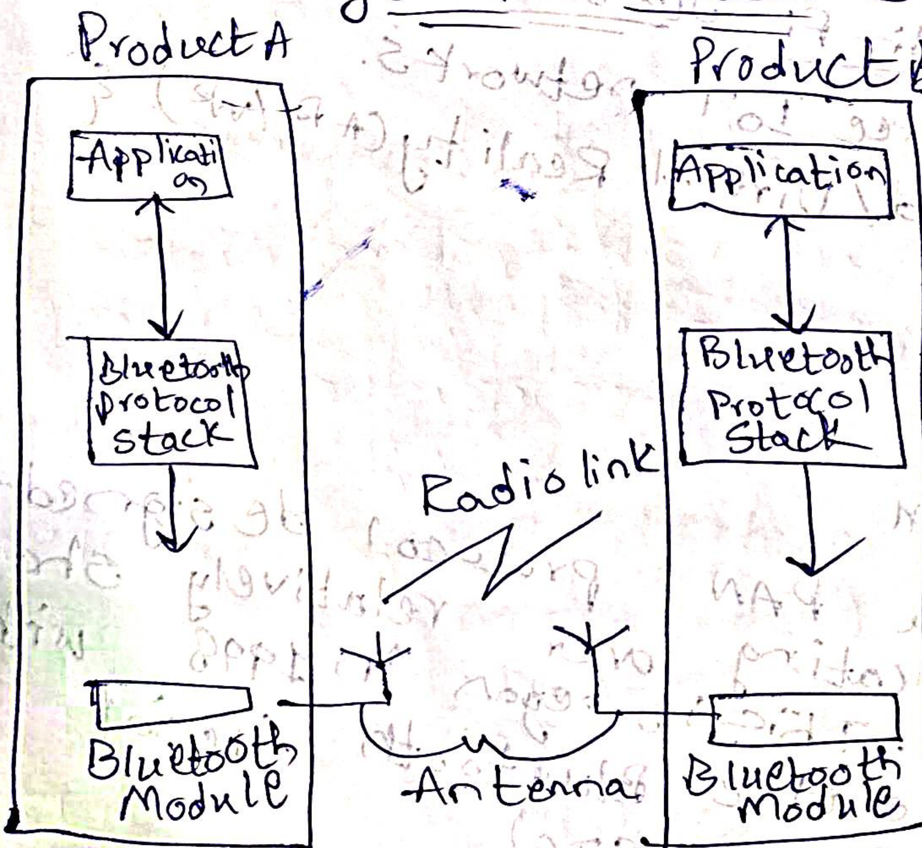
Topology

1) Point - to - Point ⇒ 2 Bluetooth devices.

2) Piconet ⇒ 8 devices

3) Scatternet ⇒ Piconets can be connected to form larger networks.

System Architecture



Protocols in the Bluetooth Protocol Stack

→ Core Protocols.
→ Cable Replacement Protocol
→ Adopted Protocol
→ Telephony Control Protocol.

Types of Bluetooth

→ Bluetooth Classic (BR/EDR)
→ Bluetooth Low Energy (LE).

* A wireless rechargeable sensor network (WRSN) is an enhanced version of WSN.
* Each sensor node uses a rechargeable battery.

Key components

→ Sensor Nodes
→ Rechargeable Batteries.
→ Wireless Energy.
→ Base Station.

Inductive Coupling

* Energy is transferred using magnetic fields between two devices.

Resonant Inductive Coupling.

* Uses tuned resonance to transfer power more efficiently over longer distances.

Internet Protocols

RIP (Routing Information Protocol)

* This is a intradomain routing protocol used inside an autonomous system.
* This is based on Distance vector routing.
* Structure is considered as a graph where nodes are the routers, and the links are the networks.
* Metric in RIP is Hop count.
  ⇓
  no of networks required to reach the destination.
* Infinity = 16. RIP should not have more than 15 hops, as a result it is used for small networks or small autonomous systems.

OSPF (Open Shortest Path First)

* This is an interior gateway protocol that has been designed within a single autonomous system.
* Based on link state routing.

OSPF messages

→ DESIGNATED ROUTER.
→ LINK STATE UPDATE
→ LINK STATE ACK
→ DATABASE DESCRIPTION.
→ LINK STATE REQUEST

BGP (Border Gateway Protocol)

* This also named as Exterior Gateway Protocol.
* Used to exchange routing information among the autonomous system on the internet.

Access Control List (ACL)

* ACL is a set of rules defined for controlling network traffic & reducing network attacks.

Types of traffic

1) Inbound access lists
   * This is applied to traffic entering an interface of a router or firewall.

2) Outbound access list
   * An outbound ACL is applied to traffic leaving an interface of a router or firewall.

Types of Access control list

1) Standard Access List
   * This filters traffic only based on source IP address.

2) Extended Access Lists
   * An Extended AC filters traffic based on,
     → Source IP
     → Destination IP
     → Port numbers.

3) Numbered Access Lists

* ACLS identified using a number.

4) Named Access Lists

* Identified using a name instead of a number.

Autonomous Systems

→ Collection of networks that are all managed, controlled & supervised by a single entity or organization.

→ Every AS's has a backbone area called Area 0. All areas are connected to backbone.

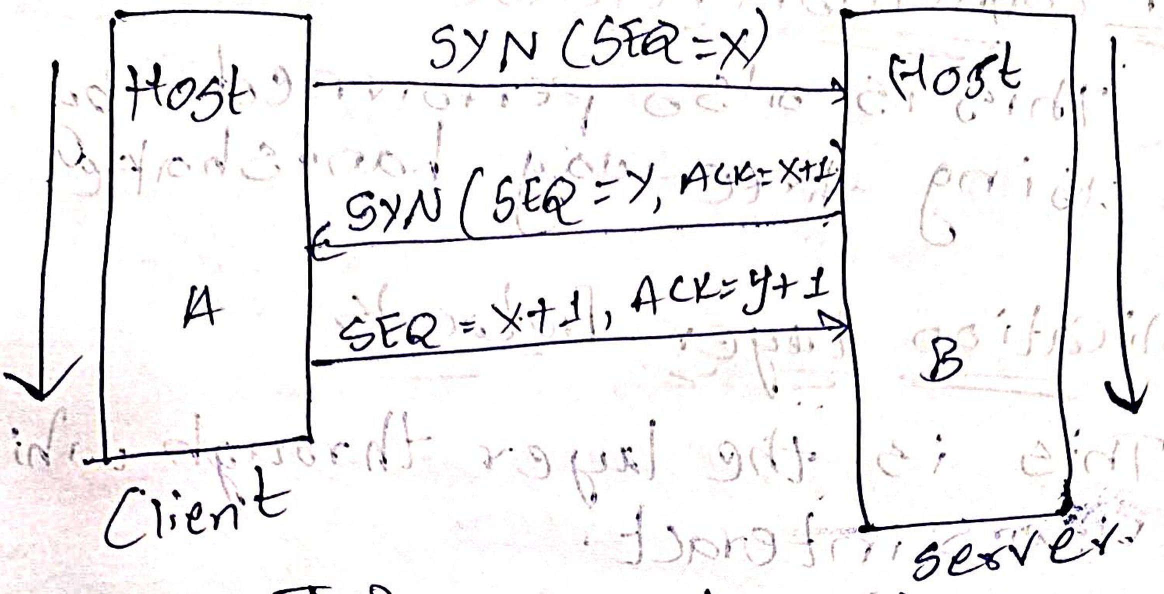
Types of Routers

→ Internal
→ Backbone
→ Area Border
→ AS Boundary

routers.

TCP Connection Establishment

* This is performed by using the three-way handshake mechanism.



Step 1 :- TCP segment ~~with the~~ sends SYN flag set to the server.

Step 2 :- Upon receiving the SYN segment from the client, the server responds with a TCP segment containing the SYN & ACK flags set.

Step 3 :- Finally, the client acknowledges the server's SYN-ACK segment by sending an ACK segment.

Note :- → SYN segment (cannot carry data) * consumes one sequence number.

→ SYN + ACK segment (cannot carry data & but does consume one sequence number.

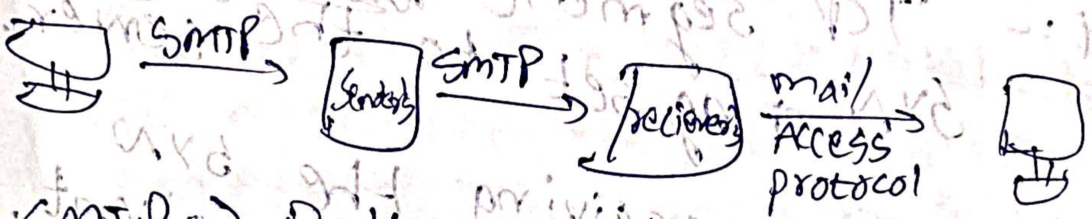
→ ACK segment ⇒ If data is not carried, no sequence number.

TCP connection Release

* This is also performed by using three-way handshake.

Application Layer Protocols

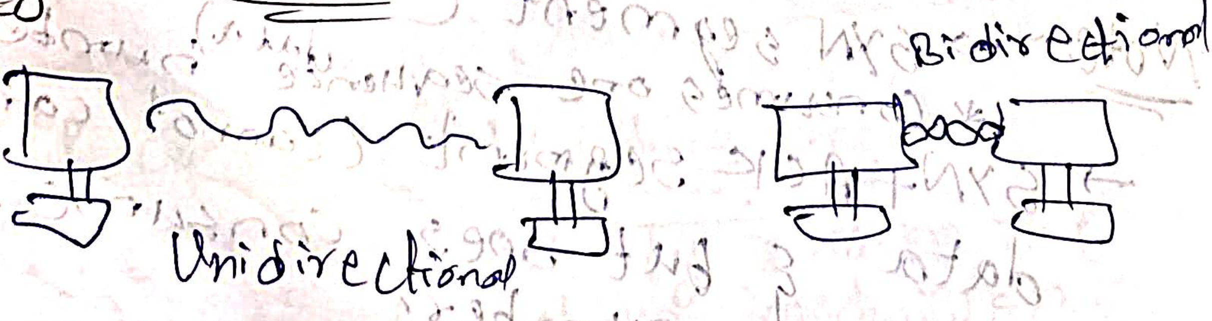
→ This is the layer through which users interact.

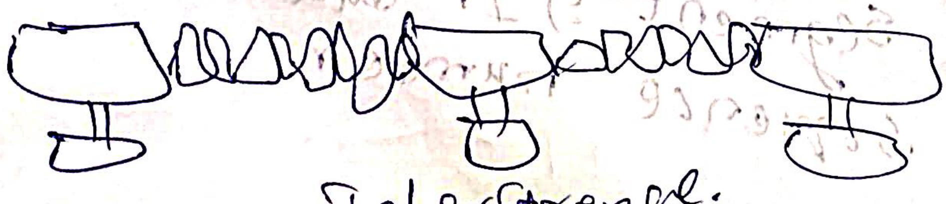


SMTP ⇒ Delivery / storage to reciever's server.
POP ⇒ Post office Protocol.
IMAP ⇒ Internet Mail Access protocol.
HTTP ⇒ gmail, Hotmail.
SMTP (Simple Mail Transfer protocol).

Wireless Communication

Key constraint





Interference.

Wireless Sensor Network (WSN):

* A wireless sensor network is a system where multiple sensore nodes are distributed across a given area to collect, monitor, & wirelessly transmit environmental data:

→ Scalability.
→ Distributed Processing
→ Autonomy.

Components

→ sensor & Sink Nodes.
→ Wirless communication.
→ Gateway.

Wireless Rechargeable Sensor Networks

* Sensore nodes are equipped with reichargeable batteries, So this ensures sustainable & continuous operation.

Cellular Technologies

* A wireless communication system that divides a large area into small cells.
* Each cell has a base station (tower).
* Enable mobile communication.

Ex :- 2G, 3G, 4G, 5G.

GSM (Global system for Mobile communications).

*A 2G cellular technology standard
*uses SIM cards to identify users.

Handoff

* The process of transfering an ongoing call or data session from one channel or base station to another
*This is initiated, when the signal from the serving base station weakens below a certain threshold & the signal of another cell is stonger.

→ Hard Handoff
→ Soft Handoff.

Cellular Capacity

* The number of users or calls that a cellular system can handle simultaneously.